CleanAir Routes — architecture

Team FAVELA · IBM Z Datathon 2026 · PM2.5 forecasting + cleanest-route guidance for Chennai

OPEN DATA SOURCES

CPCB / TNPCB monitors
Hourly PM2.5 from 7 Chennai stations + US Consulate monitor, via the OpenAQ API

Open-Meteo weather
Temperature, humidity, rain, wind — history and 48 h forecast

OpenStreetMap roads
Street network of all station areas — 120,566 junctions

India NAQI standard
CPCB breakpoints for AQI categories and health advice

IBM LinuxONE · IBM z17 (s390x) · LinuxONE Community Cloud — everything below runs here

AI PIPELINE

1 · Ingest & clean
Automatic download, sensor merging, gap filling, IST time

→

2 · Forecast PM2.5
5 horizon models (1–24 h), gradient boosting + persistence hybrid; calibrated p90 worst case

→

3 · Pollution on every street
Between-station estimate (chosen by leave-one-station-out) + busy-road uplift

→

4 · Exposure-aware routing
Inhaled dose = PM2.5 × breathing × time → fastest / cleanest / balanced + best time to leave

z17 Integrated Accelerator for AI
Neural forecaster on the on-chip NNPA via torch_nnpa — ~20× faster than CPU, same results

IBM Granite 3.3 2B (local)
Big-endian build via llama.cpp writes advice & officials' bulletin — no data leaves the machine

Secure container
Docker on IBM's LinuxONE image, auto-restart, localhost-only, reached through an encrypted SSH tunnel

WHO USES IT

Route planner (web map)
Asthma patients, elders, children, parents, commuters, cyclists: cleanest route, best time to leave, advice in English + தமிழ்

Health officials' dashboard
24 h heat table per station, city summary, ready-to-send alerts for schools, hospitals and outdoor workers, CSV export

Validated results
Beats "no change" at every horizon (winter up to 17 %, monsoon up to 21 %); worst-case band covers ~91 % of winter values up to 12 h

Python · pandas · scikit-learn · PyTorch · NetworkX · Flask · Leaflet · Chart.js · llama.cpp · Docker

Event code SSF2026